
Conestoga College
MATH73235: Advanced Topics in Mathematics
Spring 2025 Midterm 2
July 15, 2025

Exam Duration: 5pm - 7pm.

Student Name (print): Stuart Murray

Student ID: 8864228

This exam contains 10 pages (including this cover page) and 5 questions. The total number of possible points is 50. Enter your answers in the space provided.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided.
- **Mysterious or unsupported answers will not receive full credit.** Correct answers should be supported by calculations, explanations, or algebraic work to receive full credit; incorrect answers supported by substantially correct calculations and explanations may still receive partial credit.
- **Provide exact answers** unless otherwise instructed.
- **Simplify all answers as much as possible.** This means that you need to combine like terms, reduce fractions, etc.
- **Be sure to state units for applied problems.**
- **Clearly identify your answer for each problem.**

Do not write in the table to the right.

Question	Points	Score
1	10	3
2	10	10
3	10	8
4	10	10
5	10	9
Total:	50	40

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1. (10 points) **Conceptual Questions.** Write an equation, an expression, or a short sentence to describe each of these concepts.

- (a) The expression for the volume integral $\iiint_T f(x, y, z) dx dy dz$ when the coordinates are changed from (x, y, z) to some other coordinates (u, v, w) .

Solution.

$$\iiint_T f(x(u, v, w), y(u, v, w), z(u, v, w)) \frac{\partial(x, y, z)}{\partial(u, v, w)} du dv dw$$

□

- (b) The main equation for **Stokes' Theorem** for a vector field $\vec{F}$ in a surface S with a boundary curve C traversed such that S remains on the left of C .

Solution.

$$S: r \leq C$$

$$\iint_A \vec{F} \cdot \hat{n} dA$$

$$\text{Stokes: } \iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} dA = \oint_C \vec{F} \cdot d\vec{r}$$

□

- (c) An expression for a vector $\vec{N}$ that is normal to the surface $z - x^2 - y^2 = 0$.

Solution.

$$\vec{N} = \frac{z - x^2 - y^2}{\sqrt{(z - x^2 - y^2)^2}} \Rightarrow \vec{N} = \frac{z - x^2 - y^2}{\sqrt{z^2 - x^4 - y^4}}$$

$$\vec{N} = \frac{\hat{n}}{|\hat{n}|}$$

$\vec{N}$ has to be a vector. $\vec{N} = \vec{\nabla}g$ is ok

□

- (d) The vector field $\vec{F}$ polar coordinates written as $F_r \hat{e}_r + F_\theta \hat{e}_\theta$ if in Cartesian coordinates we have $\vec{F} = F_1 \hat{i} + F_2 \hat{j} = (x^2 + y^2) \hat{i} - 2y \hat{j}$. Hint: $F_r = F_1 \cos \theta + F_2 \sin \theta$, $F_\theta = -F_1 \sin \theta + F_2 \cos \theta$.

Solution.

$$\begin{aligned} & [F_1 \cos \theta + F_2 \sin \theta] \hat{e}_r + [-F_1 \sin \theta + F_2 \cos \theta] \hat{e}_\theta \\ & [(x^2 + y^2) \cos \theta - 2y \sin \theta] \hat{e}_r - [(x^2 + y^2) \sin \theta + 2y \cos \theta] \hat{e}_\theta \end{aligned}$$

□

- (e) The expression for the divergence $\nabla \cdot \vec{F}$ in spherical coordinates (ρ, θ, ϕ) if we are given that the divergence of the same vector field is $\nabla \cdot \vec{F} = r^2 + z^2$ in cylindrical coordinates (r, θ, ϕ) .

Solution.

$$\begin{aligned} & \iiint (\nabla \cdot \vec{F}) r dr d\theta dz \quad ? \quad \text{why integrate?} \\ & = \iiint [x^2 + y^2 + z^2] r dr d\theta dz = \iiint [\rho^2 \cos^2 \theta \sin^2 \phi + \rho^2 \sin^2 \theta \sin^2 \phi + \rho^2 \cos^2 \phi] \rho^2 \sin \phi d\rho d\theta d\phi \\ & = \iiint \rho^4 \sin \phi d\rho d\theta d\phi \end{aligned}$$

□

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2. (10 points) Use the appropriate Lamé coefficients to compute the Laplacian of the scalar function

$$f(x, y, z) = x^2 + y^2 + z^2$$

in both Cylindrical and Spherical coordinates. You must compute all the required partial derivatives.

Solution.

Cylindrical: $\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial r} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial z} \right) \right]$

$x = r \cos \theta$ $h_1 = 1$
 $y = r \sin \theta$ $h_2 = r$
 $z = z$ $h_3 = 1$

$$= \frac{1}{r} \left[\frac{\partial}{\partial r} (r (2r)) + \frac{\partial}{\partial \theta} \left(\frac{1}{r} \cdot 0 \right) + \frac{\partial}{\partial z} (r (2z)) \right]$$

$$= \frac{1}{r} [4r + 2r] \Rightarrow 6$$

$$F = r^2 \cos^2 \theta + r^2 \sin^2 \theta + z^2$$

$$F = r^2 + z^2$$

Spherical

$$\rho = \rho \cos \theta \sin \phi \quad h_1 = 1$$

$$y = \rho \sin \theta \sin \phi \quad h_2 = \rho \sin \phi$$

$$z = \rho \cos \phi \quad h_3 = \rho$$

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial \rho} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial \rho} \right) + \frac{\partial}{\partial \theta} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial \theta} \right) + \frac{\partial}{\partial \phi} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial \phi} \right) \right]$$

$$= \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} (\rho^2 \sin \phi (2\rho)) + \frac{\partial}{\partial \theta} \left(\frac{\rho}{\sin \phi} \cdot 0 \right) + \frac{\partial}{\partial \phi} \left(\frac{\rho \sin \phi}{\rho} \cdot 0 \right) \right]$$

$$= \frac{1}{\rho^2 \sin \phi} [6 \rho^3 \sin \phi] \Rightarrow 6$$

$$F = \rho^2 \cos^2 \theta \sin^2 \phi + \rho^2 \sin^2 \theta \sin^2 \phi + \rho^2 \cos^2 \phi$$

$$= \rho^2 [\sin^2 \phi (\cos^2 \theta + \sin^2 \theta) + \cos^2 \phi]$$

$$= \rho^2 [\sin^2 \phi + \cos^2 \phi]$$

$$= \rho^2$$

∴ Spherical = Cylindrical

$$6 = 6$$

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3. (10 points) Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

Solution.

$$\phi = \text{const} = \frac{\pi}{6}$$

$$0 \leq \rho \leq 2 \quad \checkmark$$

$$0 \leq \theta \leq 2\pi \quad \checkmark$$

$$\begin{aligned} |h_1 h_2 h_3| &= (1)(\rho \sin \phi)(\rho) \\ &= \rho^2 \sin\left(\frac{\pi}{6}\right) \\ &= \frac{1}{2} \rho^2 \end{aligned}$$

$$dA = \frac{1}{2} \rho^2 d\rho d\theta$$

$$\iint \vec{F} \cdot \hat{n} dA \Rightarrow \int_0^{2\pi} \int_0^2 \frac{1}{2} \rho^2 d\rho d\theta$$

For $\phi = \text{const}$
surface

$$dA = \rho \sin \phi d\rho d\theta$$

$$= \frac{1}{6} \int 8$$

$$= \frac{4}{3} [\theta]_0^{2\pi}$$

$$= \frac{8\pi}{3}$$

Correct calculation
of integral
(error carried)

□

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4. (10 points) Verify Gauss' Divergence Theorem by computing the following surface integral:

$$\iint_S \vec{F} \cdot \hat{n} \, dA,$$

and the volume integral:

$$\iiint_T (\nabla \cdot \vec{F}) \, dV,$$

where $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$, S is the surface of a sphere of radius 2 that forms the boundary of T .

Hint: This problem may be easier to solve in the spherical coordinates. Note that in spherical coordinates, $\vec{F} = \rho\hat{e}_\rho$, while for S , $\hat{n} = \hat{e}_\rho$, and thus

$$\vec{F} \cdot \hat{n} = (\rho\hat{e}_\rho) \cdot (\hat{e}_\rho) = \rho.$$

Solution.

$$\vec{F} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\begin{aligned} S: 0 \leq \rho \leq 2 \\ 0 \leq \theta \leq 2\pi \\ 0 \leq \phi \leq \pi \end{aligned} \quad \begin{aligned} x &= \rho \cos \theta \sin \phi \\ y &= \rho \sin \theta \sin \phi \\ z &= \rho \cos \phi \end{aligned}$$

$$\begin{aligned} dV &= |h_1 h_2 h_3| \, d\rho \, d\theta \, d\phi \\ &= \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi \end{aligned}$$

$$\begin{aligned} dA &= h_2 h_3 \, d\theta \, d\phi \\ &= \rho^2 \sin \phi \, d\theta \, d\phi \end{aligned}$$

$$\vec{F} \cdot \hat{n} = \rho \Rightarrow \iint_S \vec{F} \cdot \hat{n} \, dA \rightarrow \iint_S \rho \, dA$$

$$= \iint_S \rho^3 \sin \phi \, d\theta \, d\phi$$

$$\rho = 2 \rightarrow = 8 \iint_S \sin \phi \, d\theta \, d\phi$$

$$= 8 \int_0^{2\pi} \int_0^\pi \sin \phi \, d\phi \, d\theta$$

$$= 16\pi \int_0^\pi \sin \phi \, d\phi \rightarrow 16\pi [-\cos \phi]_0^\pi$$

$$16\pi [1 - (-1)] \Rightarrow 32\pi$$

$$(\nabla \cdot \vec{F}) = \frac{\partial}{\partial x} x + \frac{\partial}{\partial y} y + \frac{\partial}{\partial z} z \Rightarrow 3$$

$$= \iiint_T 3 \rho^3 \sin \phi \, d\rho \, d\theta \, d\phi$$

$$= \iiint_T \left[\frac{3\rho^4}{4} \sin \phi \right]_0^2 \, d\theta \, d\phi$$

limits?

$$= 8 \iint_S \sin \phi \, d\theta \, d\phi$$

$$= 8 \int_0^{2\pi} \int_0^\pi \sin \phi \, d\phi \, d\theta$$

$$= 16\pi [-\cos \phi]_0^\pi$$

$$= 16\pi + 16\pi \Rightarrow 32\pi$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} \, dA = \iiint_T (\nabla \cdot \vec{F}) \, dV$$

$$32\pi = 32\pi$$

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Continue problem 4

Solution.

□

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5. (10 points) In the first quadrant of the Cartesian plane (i.e., $x \geq 0, y \geq 0$), we may define a set of coordinates known as the hyperbolic coordinates (u, v) , with transformation function

$$x = ve^u, \quad y = ve^{-u}.$$

Find the area of the triangular region R given by

$$0 \leq v \leq u, \quad 0 \leq u \leq 3.$$

Solution.

$$F = \begin{bmatrix} ve^u \\ ve^{-u} \\ 0 \end{bmatrix}$$

$$\iint_R \vec{F} \cdot \hat{n} dA$$

$$= \iint_R 2(-2v) dv du$$

$$= -2 \int_0^3 \frac{2v^2}{2} \Big|_0^u du$$

$$= -2 \int_0^3 u^2 du$$

$$= -\frac{2}{3} [u^3]_0^3 \Rightarrow = -\frac{2}{3} [27]$$

$$= -18$$

$$\nabla \cdot \vec{F} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} = 2$$

Lamé:

$$h_1^2 = h_v^2 = \sum_{k=1}^2 \left(\frac{\partial x}{\partial v} \right)^2 + \left(\frac{\partial y}{\partial v} \right)^2$$

$$h_1^2 = e^{2u} + e^{-2u}$$

$$h_1 = e^u + e^{-u}$$

$$h_2^2 = h_u^2 = \sum_{k=1}^2 \left(\frac{\partial x}{\partial u} \right)^2 + \left(\frac{\partial y}{\partial u} \right)^2$$

$$= ve^{2u} + (-ve^{-u})^2$$

$$h_2^2 = v^2 [e^{2u} + e^{-2u}]$$

$$h_2 = v(e^u + e^{-u})$$

$$D = \begin{vmatrix} \frac{\partial x}{\partial v} & \frac{\partial x}{\partial u} \\ \frac{\partial y}{\partial v} & \frac{\partial y}{\partial u} \end{vmatrix} = (e^u(-ve^{-u})) - (ve^u(e^{-u}))$$

$$= -v - v$$

$$= -2v \quad \text{①}$$

← use absolute value

$$e^u e^{-u} = e^0 = 1$$

$$\frac{\partial(x, y)}{\partial(u, v)} = |\det D| = |-2v| = 2v$$

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Formula Sheet

General Coordinates:

$$(x_1, x_2, x_3) \rightarrow (q_1, q_2, q_3):$$

$$x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3)$$

$$\text{Derivative Matrix: } D = \begin{bmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_1}{\partial q_2} & \frac{\partial x_1}{\partial q_3} \\ \frac{\partial x_2}{\partial q_1} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_2}{\partial q_3} \\ \frac{\partial x_3}{\partial q_1} & \frac{\partial x_3}{\partial q_2} & \frac{\partial x_3}{\partial q_3} \end{bmatrix}$$

$$\text{Lamé coefficients: } h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2$$

$$\text{Jacobian: } \frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)} = |J| = |\det D| = |h_1 h_2 h_3|$$

$$\text{Line element: } ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2$$

$$\text{Volume element: } dV = h_1 h_2 h_3 dq_1 dq_2 dq_3$$

$$\text{Area element for surface with } q_j = \text{const} \ (i, j, k \text{ distinct}): dA = h_i h_k dq_i dq_k$$

$$\text{Gradient for scalar function } f: \nabla f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} \hat{e}_{q_1} + \frac{1}{h_2} \frac{\partial f}{\partial q_2} \hat{e}_{q_2} + \frac{1}{h_3} \frac{\partial f}{\partial q_3} \hat{e}_{q_3}$$

$$\text{Divergence and curl of vector field } \vec{F} = F_{q_1} \hat{e}_{q_1} + F_{q_2} \hat{e}_{q_2} + F_{q_3} \hat{e}_{q_3}:$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 F_{q_1}) + \frac{\partial}{\partial q_2} (h_3 h_1 F_{q_2}) + \frac{\partial}{\partial q_3} (h_1 h_2 F_{q_3}) \right]$$

$$\nabla \times \vec{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_{q_1} & h_2 \hat{e}_{q_2} & h_3 \hat{e}_{q_3} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_{q_1} & h_2 F_{q_2} & h_3 F_{q_3} \end{vmatrix}$$

Laplacian of scalar field f :

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

Cylindrical Coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

Spherical Coordinates:

$$x = \rho \cos \theta \sin \phi, \quad y = \rho \sin \theta \sin \phi, \quad z = \rho \cos \phi$$

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

sin
cos
-sin
-cos